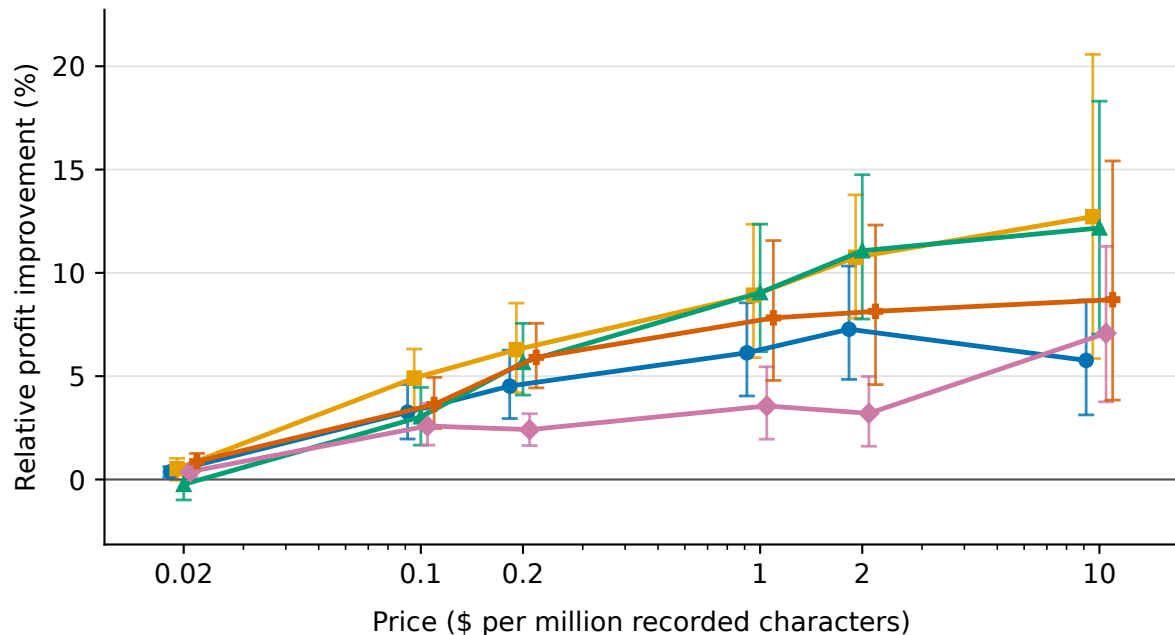


FSFairX reward model · adaptive stopping versus best-of-N

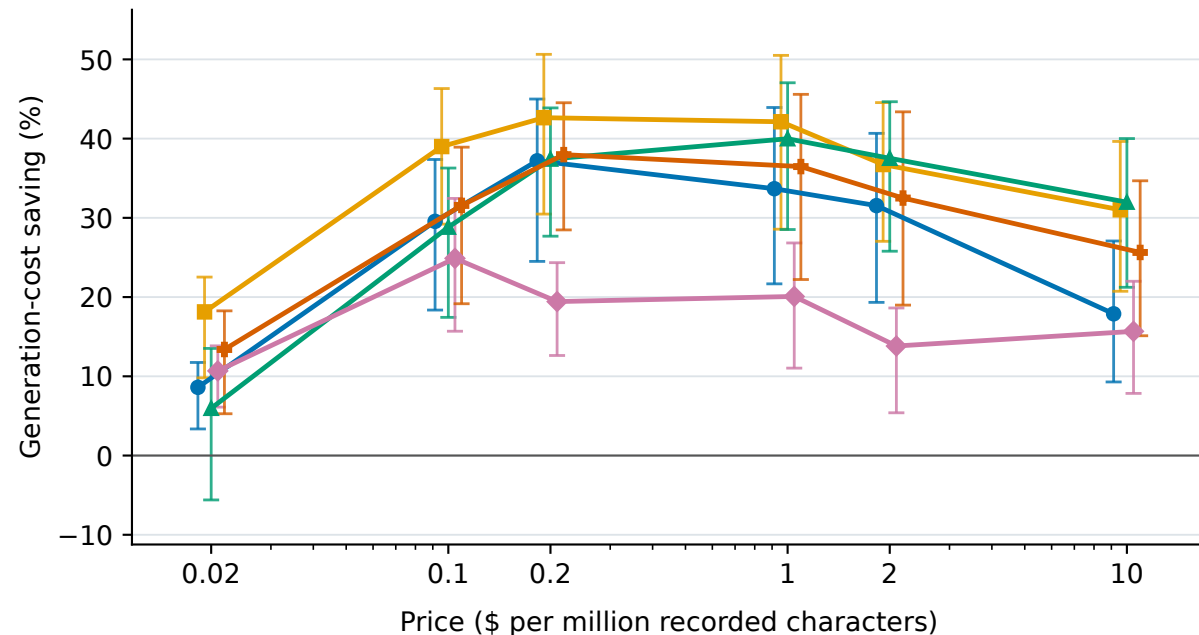
Generation model

- Gemma-2-9B
- Llama-3.1-8B
- Llama-3.2-3B
- Mistral-7B
- Qwen-2.5-7B

(a) Profit improvement



(b) Cost savings at matched quality



Whiskers: pointwise 95% prompt-bootstrap intervals (1,000 resamples); small x-offsets separate generators.

Profit baseline: training-selected fixed N. Cost baseline: retrospective fixed-N mixture matching the policy's attained quality.

Shared prompt resampling respects overlapping splits. Intervals condition on fixed policies/selections; not simultaneous or selection-adjusted.